Practical 10

Create REST API using Python Flask to send the form data like username, email into the database and test that API.

To create a REST API using Python Flask that receives form data (e.g., username and email) and stores it in a database, you can follow these steps.

Step 1: Install Flask and Flask-SQLAlchemy
If you haven't already, install Flask and Flask-SQLAlchemy:

pip install Flask Flask-SQLAlchemy

Step 2: Create Your Flask Application
Create a Python script, e.g., app.py, and build your Flask application:

```
from flask import Flask, request, jsonify
from flask_sqlalchemy import SQLAlchemy

app = Flask(__name__)

# Configure the SQLite database (you can change it to PostgreSQL)
app.config['SQLALCHEMY_DATABASE_URI'] = 'sqlite:///mydatabase.db'
db = SQLAlchemy(app)

# Define a User model
class User(db.Model):
    id = db.Column(db.Integer, primary_key=True)
    username = db.Column(db.String(80), unique=True, nullable=False)
    email = db.Column(db.String(120), unique=True, nullable=False)

    def __init__(self, username, email):
        self.username = username
        self.email = email

# Create database tables
db.create_all()

# Define a route to receive and store form data
@app.route('/api/user', methods=['POST'])
def create_user():
    if request.method == 'POST':
        data = request.json  # Assuming JSON data is sent
        username = data.get('username')
        email = data.get('email')

        if not username or not email:
            return jsonify({'message': 'Both username and email are required'}), 400

        user = User(username=username, email=email)
        db.session.add(user)
        db.session.commit()
        return jsonify({'message': 'User created successfully'}), 201

if __name__ == '__main__':
    app.run(debug=True)
```

Step 3: Test Your API

You can use a tool like **curl** or Postman to test your API.

Here's an example using **curl** to send a POST request to create a new user:

curl -X POST -H "Content-Type: application/json" -d '{"username": "rahul_patel", "email": "rahul@example.com"}' http://localhost:5000/api/user

This command sends a JSON payload to the **/api/user** endpoint, creating a new user in the database.